

ASCII MASTER FLOW – PANEL 1/12: Desert identity test

MOISTURE DEFICIT -> desert criterion

HOT + ARID -> BWh tendency

COLD WINTER + ARID -> BWk tendency

SAND / HEAT -> possible, not defining

MUST REMEMBER: Hot deserts cluster near subtropical subsidence, western-margin cold currents, continental interiors and rain shadows, while mid-latitude deserts add distance and mountain barriers; large diurnal range, episodic rain and xerophytes follow water-energy balance.

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ASCII MASTER FLOW – PANEL 2/12: Aridity cause matrix

SUBSIDENCE -> stable descending air

OFFSHORE FLOW -> weak moisture supply

COLD CURRENT -> fog but weak convection

CONTINENT / RAIN SHADOW -> inland dryness

ASCII MASTER FLOW – PANEL 3/12: Cold-current coast

COLD WATER -> chilled marine boundary layer

STABLE AIR -> weak vertical uplift

FOG / MIST -> moisture without deep rain

COASTAL DESERT -> aridity persists

ASCII MASTER FLOW – PANEL 4/12: Temperature firewall

HOT DESERT -> very large day-night range

MID-LATITUDE DESERT -> cold winter

CLEAR SKY -> rapid nocturnal radiation

RULE -> do not equate desert with constant heat

ASCII MASTER FLOW – PANEL 5/12: Wadi flood chain

RARE INTENSE STORM -> high rain rate

BARE / CRUSTED SURFACE -> rapid runoff

DRY CHANNEL -> sudden flow concentration

SETTLEMENT / ROAD -> flash-flood exposure

ASCII MASTER FLOW – PANEL 6/12: Plant-water adaptations

LEAF -> reduced, waxy or thorny

STEM -> water storage in some species

ROOT -> deep tap or wide shallow network

SPACING -> lower competition for water

ASCII MASTER FLOW – PANEL 7/12: Oasis and water strategy

AQUIFER / SPRING / EXOTIC RIVER -> source

WELL / CONDUIT / TANK -> access

DATE PALM + CROPS -> layered use

OVERUSE -> depletion, salinity or conflict

ASCII MASTER FLOW – PANEL 8/12: Thar aridity map

ARABIAN SEA BRANCH -> approaches north-west

ARAVALLIS -> broadly parallel orientation

WEAK FORCED ASCENT -> little orographic rain

CONTINENTALITY + SUBSIDENCE -> reinforce aridity

ASCII MASTER FLOW – PANEL 9/12: Thar landscape ledger

DUNES -> barchan, seif and sand sheets

DHANDS / PLAYAS -> inland salt basins

LUNI -> principal inland-draining river

KHEJRI / GRASS -> xeric open habitat

ASCII MASTER FLOW – PANEL 10/12: Transformation trade-off

TANKA / KHADIN / JOHAD -> local harvesting

CANAL -> irrigation and settlement expansion

RENEWABLE GRID -> low-carbon power

RISKS -> salinity, waterlogging, habitat collision

CLOSE DISTINCTION: Hot desert differs from cold/mid-latitude desert, aridity from drought, and cold current promotes coastal stability/fog rather than directly 'removing' all rain; Thar's monsoon-margin setting differs from Sahara's dominant controls.

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ASCII MASTER FLOW – PANEL 11/12: UNCCD status gate

COP16 -> Riyadh, 2-13 December 2024

LAND + DROUGHT -> resilience agenda advanced

BINDING FRAMEWORK -> not finalised

PYQ LEDGER -> no direct Topic 18 demand

EVIDENCE LIMIT: Desert location is multicausal and boundaries migrate; Great Indian Bustard habitat policy, renewable infrastructure and desertification trends require dated, spatially explicit sources and trade-off analysis.

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ASCII MASTER FLOW – PANEL 12/12: Desert answer spine

DEFINE -> aridity and water balance

MAP -> circulation, current and relief causes

COMPARE -> hot versus mid-latitude

APPLY -> Thar, GIB and drought governance